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The Anthropocene, Climate Change and (Ecological) Security

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Introduction

Security is a master concept of IR thought and practice. The promise of providing it underpins the legitimacy of key institutions of global politics, whether states or international organizations. For the former, the commitment to provide security is central to the social contract. For the latter, the commitment to the 'maintenance of international peace and security' is featured as Article 1 of the United Nations (UN) Charter. And for prominent theories of international relations, the quest for security drives key dynamics of global politics and attempts to govern or regulate it. But to what extent do either the practices and governance of security in global politics, or theories of security in IR thought sufficiently come to terms with the scale of contemporary ecological crises or the broader Anthropocene context in which these are embedded? Does the embrace of climate change as a security issue in states or international organizations point to a (necessary) shift in the ways in which climate change, security and/or global governance are conceived or approached?

This chapter explores these questions, pointing in the process to the problematic logics that continue to inhere in prominent approaches to 'climate security' even when this link is acknowledged or embraced. Of course, that there are limits to existing global practices and governance arrangements in effectively coming to terms with the realities of climate change is a conclusion that may be drawn on the basis of the onward march of the realities of climate change, species extinction and unprecedented (in the human experience at least) challenges to ecosystems and their resilience. But simply pointing to the failure of states and the broader 'international

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community' to address a challenge like climate change in the 40 years in which we have been aware of the problem doesn't get us to the reasons for this failure or means of addressing it. Rather, coming to terms with this failure requires examining the core assumptions and logics underpinning IR thought and global practice regarding the Anthropocene and ecological crises. Here, the assumptions, logics and pathologies of 'security' provide a particularly important insight into this failure, not least as a central concept in IR thought.

Ultimately, as this chapter will explore, even when we have seen recognition of the threats posed by climate change, this has stopped well short of enabling or ushering in the types of measures or practices necessary for genuinely addressing the harms associated with it. The engagement of the UN Security Council (UNSC) with climate change, for example, has increased over time and states involved in these deliberations are eager to affirm the importance of action to address the challenges posed by climate change. Yet a resolution recognizing the UNSC's responsibilities to contribute to addressing this threat – even involving minimal commitments such as regular reporting to the Secretary-General or the integration of climate considerations in the design and execution of peace operations, for example – has proven to be beyond the capacity or willingness of the institution or its member states. In this sense, and as contributors to this debate argued at the time (see McDonald 2023b), the capacity of the institution to claim to be fulfilling its role as the primary institution for 'maintaining international peace and security' – central to the UN mandate – could be called into question. Speaking to the core concerns of this volume, it is certainly possible to conclude that the inherent limitations of attempts to govern climate change responses through the lenses and institutions of security- for reasons that will be examined- risk further undermining global order and the legitimacy of key institutions within it.

In simple terms, the central idea or image of security evident in both theory and modes of security governance- that of self-contained units attempting to insulate themselves from deliberate external threats- is simply inimical to coming to terms with the challenge of an issue like climate change. In the context of climate change the origins of the 'threat' are predominantly unintended (and everyday) action both within and beyond the state; the state lacks the capacity to unilaterally address that threat regardless of its material power, capacity or contribution to the problem; the traditional instruments of (national) security are either largely marginal to the means of responding to it or even a significant part of the problem (see, for example, Crawford 2022); the idea of insulation or conservation of self-contained units does not get to grips with the reality of ongoing (and unavoidable) change to ecosystem functionality and with it life prospects; the necessity for cooperation between states immediately runs into complex ethical

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challenges and problematic governance systems, structures and norms; and, perhaps most fundamentally, the separation of humanity and nature central to conceptions of what it means to be secure – admittedly dominant in modern political thought (see Soper 1995; Clark and Szerszynski 2021) – is fundamentally at odds with the need to come to terms with the reality of the Anthropocene context and the idea of humanity as irrevocably embedded in the ecological contexts upon which it relies for its ongoing existence. All these points suggest fundamental impediments to addressing ecological crises through existing institutions and practices. They also suggest fundamental impediments to understanding the origins of these crises, or guiding responses to them, through the dominant frameworks of traditional IR thought. These limitations become evident when examining current institutional responses to the climate change-security relationship, and the international relations discourses with which these responses align.

This chapter is divided into three sections. In the first section, I outline engagement with security in IR thought and practice. I identify key assumptions regarding the meaning and logic of security in IR thought before examining what the practice or governance of security looks like. The second section outlines the challenge of the Anthropocene context and ecological crises to this image of security, focusing in particular on the challenge of climate change. After noting the scale of this challenge, I examine attempts in IR thought and practice to come to terms with the security implications of climate change. The third and final section reflects on where this analysis leaves us, pointing to the promise of approaching the security implications of ecological crises through the lens of ecological security, in which our focus turns to the resilience of ecosystems themselves. I situate this approach in broader attempts to challenge the human-nature divide, before noting key challenges facing this approach and its institutionalization that speak to the core concerns of this volume as a whole.

Security in international relations

Security is a master concept in IR thought and practice. Of course, for traditional approaches to international relations such as realism, the (endless) search for security – defined principally in terms of the survival of the state – is the driving force of IR. This has always been a moral project, as Michael Williams (1998) has argued. At the heart of the realist defence of the primacy of the state is the social contract: the idea that individuals sacrifice some degree of freedom and autonomy to the state – the Leviathan, in Hobbes' terms – in return for their protection in an anarchic and dangerous external environment. A concern with states and their security is, then, a concern with the welfare of people. Viewed in this light, it becomes easier to understand why state leaders consistently emphasize that the promise of

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providing security is the most important responsibility of the government itself. Perhaps more importantly for the purposes of this chapter and the broader text of which it is a part, recognizing this point gives us a sense of the centrality of security to IR thought and practice.

In the national security framework,[1] which is still dominant in IR thought and practice in the face of sustained critique from scholars in the 'critical security studies' tradition (see Browning and McDonald 2013), security is of – and for – states. It is the state whose security – defined in terms of sovereignty and territorial integrity – matters. And it is the state, through self-help and military preparedness in particular, that provides security. Building on its realist foundations, the imperative of self-help in a world characterized by the absence of a higher authority than states encourages the aggressive and single-minded pursuit of material power relative to others to enable the state's survival, and further facilitate the state's capacity to ensure that survival through the projection of power externally.[2]

The alignment of referent and agent of security in this framework provides a neat answer to key questions of ethical focus and responsibility, with the state positioned as being concerned with protecting itself. However, this necessarily positions the space of security practice as one in which self-contained units (states) attempt to insulate themselves from external threats to their sovereignty and territorial integrity. By extension, issues become security issues to the extent that they have the capacity to threaten or undermine sovereignty or territorial integrity. This establishes relatively clear criteria for considering whether issues of environmental change are considered on the national security agenda, but also encourages a highly partial view of the nature of the challenge posed by environmental change. While climate change causes significant and widespread harm to peoples and ecosystems, fundamentally undermines conditions for human survival and even challenges the capacity of states to function, it does not follow that – within the national security framework – it warrants consideration as a national security threat (Levy 1995). I will return to this point later in this chapter.

In the international security framework, in which the focus turns to the stability of the international system and/or the protection of the rules and norms of an international society (see Bellamy and McDonald 2004), there is clearly more room for states – still the most powerful actors – to work together to address threats to security. This capacity to recognize 'absolute' rather than simply relative gains allows for the possibility of addressing transnational threats, which is clearly more consistent with recognizing and responding to environmental change (see Garcia 2010). However, the question of referent object and agent of security becomes significantly more complicated. As critics have rightly noted, abstract notions of an 'international community' in need of either defending or acting appear at

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once both unclear and partial in their account of whose community matters or exactly who is responsible for acting (Buzan and Gonzalez-Pelaez 2005).

A more productive account of what international security means in IR theory might here draw on the English School focus on international society (Dunne 1998; Bellamy and McDonald 2004). Within this international society, we see a shifting emphasis (in terms of what constitute dominant norms and rules) between concerns with 'order' (pluralism) and 'justice' (solidarism). While the former orients towards the maintenance of the status quo and absence of disruption to that order, the latter focuses our attention on potentially responding to harms regardless of their effect on international stability (enabling practices that challenge sovereignty as non-intervention, such as humanitarian intervention). There is an implication that states can and should work together to address either the disruptive implications of the effects of environmental change that threaten international stability or to address the harmful effects of environmental change through both prevention and response. As such, states remain the most powerful agents, but recognize the need to work together in concert with other actors to address challenges like climate change.

In the national security framework, which is embedded in realist thought, the pathway for an issue like climate change to be considered a national security threat is limited (Levy 1995). If the requirement is the potential threat posed to sovereignty and territorial integrity, few states experience such considerations directly as a result of climate change. At best we might acknowledge the potential immediate threat posed to low-lying Pacific Island states, for example, whose territory (even sovereignty by some accounts) is immediately and directly threatened by rising sea levels associated with the effects of climate change (Maas and Carius 2012; Mitra and Sanghi 2023). But this scenario is rare, and for the most part our focus in the national security framing is with the secondary implications of climate change for national security: the threat posed by population movements or conflict itself linked to or caused by climate change (Busby 2007, 2008). Indeed, this is at the heart of the 'threat multiplier' (CNA 2007, 2014) framing that has become dominant in accounts of the national security implications of climate change and its potential relationship to armed conflict. Rather than a threat in and of itself, climate change (despite the existential challenge it demonstrably represents) becomes a threat only when linked to the potential for it to trigger armed conflict. The state responds to these indirect challenges in turn not primarily through prevention (mitigation), but by addressing the consequences of this challenge through adaptation (McDonald 2013). As will be noted in the following section, while we see variations in different states' engagement with the (national) security implications of climate change, in practice this framework is evident in the way in which states conceive of the key dimensions of threat and response.

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In the case of an international security framework, the nature of the threat posed by climate change is still predominantly an indirect one: the potential for the direct effects of climate change (warmer temperatures, changing rainfall patterns, desertification, ocean acidification or disasters) to precipitate *genuine* threats associated with large-scale population movement, disruptive state fragility, regional instability or armed conflict. This is arguably the lowest common denominator vision of an international security focus – the absence of large-scale and disruptive instability. Indeed, it is significant to note here that the 'threat multiplier' framework has been invoked by a range of international organizations and in the context of debates within the UN Security Council about the international security implications of climate change (Hardt 2021; Maertens 2021; Scartozzi 2022; McDonald 2023b). There is a potential here – evident in the UN Security Council debates noted previously – for states and international organizations to push in the direction of a focus on harm, potentially emphasizing the role of mitigation action as a means of advancing security. Yet, as will be noted later on, such conceptualizations remain marginal to institutional engagement with the security implications of climate change, whether at the national or the international level.

Thus, in traditional accounts of security in IR, the story is ultimately one of key institutions of global politics attempting to insulate themselves from the destabilizing effects of external developments. In the dominant national security account, states are concerned with protecting their own sovereignty and territorial integrity from these external threats. In the case of international security states might be capable of working together to advance or address (transnational) threats that pose a challenge to international order and stability, but the threat remains largely indirect, practices orient towards the responsive (that is, adaptation), and responsibility for action depends in this schema on the extent to which we see institutional recognition of a threat (McDonald 2024a). Neither of these frameworks appears to be well suited to making sense of or guiding responses to the nature of ecological crises we face in the Anthropocene epoch. And the disinclination of key institutions to move beyond these dominant accounts risks undermining their own legitimacy in the face of the onset of climate change and, with it, potentially undermining the broader global order.

Security, the ecological challenge and the Anthropocene

It is difficult to do justice to the scale of contemporary ecological crises, memorably characterized in Chapter 1 of this volume as 'omni-crises'. Recent prominent attempts to identify and track challenges to 'planetary boundaries', for example, demonstrate a consistent failure to address

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biosphere integrity, chemical pollution, ocean acidification, freshwater availability, land system change, nitrogen and phosphorous flows to oceans and the biosphere, atmospheric aerosol loading and, of course, climate change (Rockström et al 2009, 2023). The impact of humanity on the natural world is such that 'human-made mass now exceeds all living biomass' (Elhacham et al 2020). According to a range of accounts, we are experiencing a 'sixth mass extinction' event – a 'biological annihilation' (Ceballos, Ehrlich and Dirzo 2017) – with a recent report suggesting a 70 per cent decline in wildlife populations in the last 50 years (WWF 2022). And then there's climate change.

While often presented as a pressing future challenge, the reality of climate change and its effects have shifted to the here and now. The most recent IPCC (2022) report noted that average global temperatures were already 1.1°C over pre-industrial levels, with consistent spikes over the Paris target of 1.5°C evident over the course of 2023. The climate change already experienced has resulted in more frequent and more severe disasters, species loss, changing rainfall patterns, desertification, rising sea levels and ocean acidification. This has already had devastating effects on ecosystems, some of which (for example, coral reefs) are vulnerable to relatively low-level temperature rises. With the goal of preventing dangerous warming looking less attainable, international discussions about the response to the climate crisis have shifted. From an early exclusive focus on mitigation, topics and practices such as adaptation, resilience building, reparations and even geoengineering have joined mitigation as topics in formal international climate deliberations (Bulkeley and Newell 2023; McDonald 2023a).

In making sense of these varied harms and the impact of humanity as a force on the planet, a range of analysts have made a case for the arrival of the Anthropocene epoch. Paul Crutzen (2006: 16) argued that given 'major and still growing impacts of human activities on earth and atmosphere, and at all, including global, scales, it thus is more than appropriate to emphasise the central role of mankind in geology and ecology by using the term "Anthropocene" for the current geological epoch'.

The argument for the Anthropocene followed the logic that because human activities had so fundamentally affected the functionality of Earth systems, the central contribution of humanity to the contemporary geological epoch should be acknowledged (see also Steffen 2021). In this schema, humanity, simply put, is a planetary force profoundly affecting the conditions of our own existence.

The concept of the Anthropocene has been subject to critique for eliding crucial differences regarding *which* humans have driven ecological destruction or misrepresenting the particular set of forces that have been behind this process. Indeed some grounds of critique are noted in the Chapter 1 of this volume. Jairus Grove (2019) makes a case for the Eurocene in locating

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the origins of crisis in a European model of extraction and development, while Jason Moore (2016) makes a case for the 'capitalocene' in attributing responsibility to the pursuit of particular modes of economic exchange.[3] However, for these potential limitations, this concept does draw our attention to the impossibility of separating humanity from the ecological conditions of its existence. Ultimately, it is a framing that represents a fundamental challenge to the human-nature distinction that is viewed as central to modern thought (Soper 1995). Clark and Szerzynski (2021) describe this challenge as a fundamental one to the social sciences more broadly. So just how well equipped is IR as a discipline, and the institutions of global politics themselves, for coming to terms with this challenge and guiding effective responses to it? At least in the context of more traditional approaches to security focused on the preservation of national or international security, and assessed against the responses of key institutions to this challenge, the answer is not particularly well.

The particular challenges posed by climate change, and especially the necessity of particular forms of response, clearly sit uneasily with the dominant logics of security in IR thought and extant institutional arrangements and practices in global politics. Returning to the themes outlined at the start of this chapter, the specific *nature of the threat* climate change poses is the first fundamental challenge to note. The threat is clearly an existential one, with the capacity to undermine the long-term existence of life on the planet. Yet its origins in everyday and (predominantly) unintentional actions are inconsistent with the image of the origins of threat in intentional and deliberate action by an external adversary (see Walt 2021). This also speaks to the challenge of the Anthropocene context to modern political thought and, in particular, the idea of humanity as irrevocably embedded within the ecological contexts upon which it relies for its survival. Again, an 'insulation or protection from' model of security positions the threat in terms that are inconsistent with its essence, while also reaffirming a problematic conception of who it is that is to be secured from that threat. Indeed, for some, the logic of security – of protecting or conserving the status quo, whether at the domestic or the international level – clearly pushes against the reality of ongoing change that climate change represents (see Dalby 2015; Hardt et al 2024).

Following this point, the conception of agency and means of providing security in traditional accounts of security in IR thought, and dominant practices of international politics, is also a key limitation. States are clearly still imbued with primary responsibility for addressing climate change within the UNFCCC process, and are more broadly viewed as the key security actors in global politics, with a monopoly on the legitimate use of force. Yet, demonstrably, states lack the capacity to unilaterally address the threat posed by climate change regardless of their material power, capacity or contribution

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to the problem. And here the traditional instruments of (national) security – the military and defence sectors – are either of limited immediate utility in addressing the problem of climate change itself or actively contribute to it through their own emissions (see, for example, Crawford 2022; Depledge 2023; Vogler 2024).

Finally, the need for effective global cooperation to address a global challenge clearly runs into challenges in the form of dominant modes of governance and, as noted earlier, the challenge of facilitating and pursuing meaningful action cooperatively in a world of states. The range of acutely complex ethical and political challenges in formulating a fair agreement between almost 200 states clearly suggests that the contemporary state-based order would not be a starting point of choice for managing a genuinely global challenge like climate change. These include different vulnerabilities, capacity to address implications or adapt; reliance on or profit from the fossil fuel economy; cultural and social investment in vulnerable spaces or destructive activities; historical responsibility; and different political systems, among other things (see Gardiner 2013; Bulkeley and Newell 2023). And that's before considering obligations to other living beings or future generations. The failure of the UNFCCC process, established in 1992, to prevent dangerous climate change points directly to these limitations. And while we have seen increasing recognition of the security implications of climate change among states, this has generally not encouraged the type of reorientation of perspective or new practices consistent with addressing the scale of the threat faced.

Debates within the UN Security Council seem to bear out the limitations of IR frameworks and their application to contemporary global politics. Certainly, debates about climate change within the UNSC have been increasing in frequency since 2018 and with majority support for recognizing a role for the UNSC in addressing the security implications of climate change. But substantive suggestions/proposals have been limited (to information gathering and incorporation of climate considerations into peace operations, for example) and even then we saw the rejection of this resolution (with Russia using its veto power and both China and India expressing opposition) when formally put to the UNSC in December 2021 (see McDonald 2023b). For an institution that is widely viewed as bearing primary responsibility for the provision and maintenance of international peace and security – central to its mandate – the failure to accept responsibility to act as agents for the provision of security in the case of fundamental and existential threat suggests significant limitations to existing security governance structures in global politics.[4] Something isn't working.

These limitations are evident in practical attempts to recognize, grapple with and institutionalize responses to the security implications of climate change within states. A significant majority of states now recognize the

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security implications of climate change (see Vogler 2023, 2024), but even among those states that have moved to recognize and institutionalize a position on the security implications of climate change, the same problematic pathologies, logics and implications largely remain. In terms of the nature of the threat and the definition of the referent object of security, and with a handful of exceptions,[5] the dominant focus in policy statements is on the indirect or secondary implications of climate change for sovereignty, territorial integrity and the defence sector itself (see Hardt et al 2023). Following this, we see more overarching focus in such documents on the *external* sources of threat or challenge to nation states, including the potential for militaries to be drawn into new sites of conflict or humanitarian assistance and disaster relief (HADR) missions. And very few states actually recognize the importance of mitigation, with militaries – central agents of national security within states – usually exempted from government-wide emissions reduction targets (see Depledge 2023). In all of these senses, the institutions and practices of global politics most regularly adhere to the expectations of traditional theories of IR, militating against the type of change in perspective and approach necessary for coming to terms with the Anthropocene context and the scale of contemporary ecological crises. To return to the theme of this chapter – and the broader text – the inclination here is towards attempting to fit such crises within existing (security) governance arrangements.

Australia's recent (and belated) embrace of climate change as a security concern is illustrative here. In its most recent Defence Strategic Review (DoD 2023), for example, climate change was recognized as a threat to national security, which was hardly surprising given the exposure of Australia to disasters (such as bushfires) and the acute vulnerability of small island states in its Pacific region (see McDonald 2021). Yet here, discussion of climate change warranted barely one of 100 pages in this report; most of this content was making a case for why the military should be called upon *less* to respond to natural disasters associated with climate change, and while committing no additional funds or capacity to addressing climate change, the report proceeded to make a case for a multibillion dollar investment in nuclear-powered submarines to address traditional geostrategic rivalry concerns (Evans 2024). It is tempting to conclude that climate change was little more than a footnote to a business-as-usual national security statement, crowded out by other pressing threats or concerns. Arguably, however, the issue evident here is even more fundamental: a failure of the state to genuinely reckon with the nature of the challenge posed by climate change, or to reorient conceptions, priorities, institutions and practices in a manner that is consistent with addressing this existential threat. While the nature of the climate crisis should encourage a radical reorientation of global governance and practices, as this volume suggests, the approach of key institutions to the climate-security relationship suggests a determination to ensure climate

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change – at best – fits within comfortable accounts of self-contained entities insulating themselves from external threat (see Hardt et al 2024).

A way forward?

Given these evident limitations of dominant accounts and practices of security in IR, it is hardly surprising that many advise avoiding the security framing altogether. Some see a danger of militarization of environmental challenges through this framing (Deudney 1991), others simply view the logic of security as illiberal and pernicious (Wæver 1995), while for others, a security framing suggests conservation or protection rather than recognition of the need to come to terms with ongoing (environmental) change (Dalby 2015). It's hard to wholly reject these points; indeed, some have featured in objections raised by states to the discussion of climate change in the UNSC (see Maertens 2021). But as the contributors to this volume and an increasing volume of critical scholarship notes (see Burke et al 2016), it's harder to escape the conclusion that these limitations are evident in broader IR thought and practice as a whole. Here, the veneration of states and the state system, the liberal model of political economy and development, and the primacy of an image of security defined in terms of the absence of immediate threats to the sovereignty and territorial integrity of states all militate against effective action in response to ecological crises, and at worst serve to drive it. And if security is politically significant in defining core values, in underpinning the legitimacy of the key actors of global politics and/or in enabling particular (even exceptional) forms of response (see Browning and McDonald 2013), then it needs to be redefined and reformed rather than rejected (see Booth 2007).

There have been a range of attempts in IR thought to challenge traditional images of security associated with the preservation or conservation of (self-contained) institutions in the face of deliberate external threats (see Browning and McDonald 2013). While often questioning the ethical assumptions and implications of these choices, critical scholarship on security also draws our attention to the politics and significance of apparently abstract debates about whose security matters or what constitutes a threat to that referent object. Recognizing that the promise of providing security is central to the legitimacy of key institutions of global politics, even potentially enabling particular sets of responses to challenges designated as a threat (see Wæver 1995; Buzan et al 1998), illustrates what is at stake in debates about the meaning and scope of security. In the process, it points to the intersection between IR theory and the practices of global politics that both follow from and inform the conceptual frameworks we employ.

While a range of scholarship challenges traditional accounts of security – their assumptions and implications – on a variety of grounds, more specifically

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a number of accounts have engaged with the particular need to come to terms with the nature of ecological crises. Early articulations of the need to recognize the security implications of environmental change beyond the arbitrary (and unhelpful) borders of nation states invoked a 'one world' image of planetary or biosphere security (Myers 1993; Litfin 1999). More recently, some have developed and defended the concept of 'Anthropocene security' to recognize the embeddedness of human populations in ecological contexts (Harrington and Shearing 2017; Mobjörk and Lövbrand 2021). Others have challenged the human-nature divide through articulating defences of worldly security (Mitchell 2014), posthuman security (Cudworth and Hobden 2017) or ecospheric security (Welsh 2022), for example. But perhaps the most explicit articulation of an alternative framework for making sense of the security implications of climate change in particular is that of ecological security. While noting earlier invocations of ecological security (see, for example, Mische 1989; Barnett 2001; Pirages and Cousins 2005), my own elucidation of the concept (see McDonald 2021) defines it in terms of the resilience of ecosystems themselves in the face of the immediate and direct implications of climate change.

A number of elements of this approach to the climate-security relationship are important to draw out here as a response to the tendencies and limitations of traditional approaches to and practices of security in IR. First, on the referent object, the focus on ecosystems clearly shifts our emphasis away from the (moral) primacy of existing political institutions and towards the ecological. This orientation is viewed as the best means of encouraging sets of practices that prioritize the rights and needs of the most vulnerable. The emphasis on (ecosystem) resilience – rather than conservation or preservation – reflects the reality of ongoing (environmental) change, in turn reminding us of the origins of the resilience concept in ecological thought (see Bourbeau 2018). Second, on the nature of the threat, the focus here is on the immediate and direct harms caused by climate change – rather than secondary or indirect implications – to ecosystems and, by extension, the most vulnerable. Third, on means of response, the emphasis on direct and immediate harms associated with climate change in turn involves prioritizing urgent mitigation effort to address these harms. And finally, on agents or responsibility for addressing the security implications of climate change, those with capacity (to cause and/or address harms) are defined here as bearing responsibility to act on behalf of others (on these points, see McDonald 2021).

Such an approach clearly challenges traditional accounts of security – and security threats in the context of the Anthropocene and ecological crises – in profound ways. Rather than the image of self-contained units insulating themselves from (external and deliberate) threats, it suggests the need to orient towards recognizing the embeddedness of humanity in the ecological

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conditions of its own existence and the imperative of working to minimize (direct) harms caused by challenges like climate change. States – particularly wealthy states – bear responsibility for acting to advance ecological security through mitigation action, but are not the only actors obligated to act and are far from the focal point of moral concern in terms of climate impacts. It is a significant distance, in short, from dominant accounts and practices of security in IR.

However, this distance represents both a necessary correction and a significant (potential) weakness. While the case for the former has been made, the latter concerns its limited immediate purchase or resonance within corridors of power. The elegant simplicity of a national security framing – where the state is the agent of security providing its own security – provides immediate and clear buy-in (through self-interest) from states in terms of the imperative of acting to achieve security. This is clearly absent in the case of ecological security, wherein states are compelled to act for others, from vulnerable populations in other countries to future generations of humans and other living beings. Can we realistically expect such considerations to inform the way in which security is viewed by powerful actors in global politics? Should we be surprised that something approaching 'business-as-usual' security politics appears most common?

The record to date in the embrace of ecological security among states falls a long way short of providing us with cause for optimism. This is a core point in the comparative analysis of approaches to the climate-security relationship by member states of the UNSC, in which an ecological security focus is found to be distinctly marginal to such approaches (Hardt et al eds. 2023). Yet we do see movement in this direction in terms of state engagement, with some states compelling the military and defence sector towards mitigation (NZ MoD 2019) and many identifying the imperative of meaningful and concerted international action to (urgent) mitigation action. Indeed, in his analysis of the UK's approach to the relationship between climate change and security, Cameron Harrington (2023: 315) suggests that 'there are signs that an ecological security discourse is emerging that challenges the historic focus on climate change as an accelerator of traditional security threats, or as a threat to individual well-being in vulnerable states of the Global South'.

A key challenge in this context, for scholars of climate and security and in particular for advocates of ecological security, will be identifying the circumstances or contexts that might give rise to such an approach. Here, identifying and examining movement in the direction of ecological security – recognizing progress – is important in terms of allowing us to understand the potential for movement towards practices oriented towards the most vulnerable in the face of the climate crisis (see McDonald 2021: 188–191). Interviews I conducted with policy makers in states as distinct as the US, Germany, New Zealand and Sweden in 2022–2023 suggests core concerns,

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at least at particular times and for particular governments, with preventing immediate climate harms and addressing human security concerns in response to climate change (see McDonald 2024b). Significant drivers of this type of orientation that were identified by interview subjects included public expectations, the values and commitments of particular governments and the foreign policy identity of the state concerned (see McDonald 2024b).

Beyond this, and speaking to the core concerns of this volume, it may also be the case that addressing the immediate and direct harms (insecurity) of climate change necessitates and/or drives new institutional arrangements and modes of governance. Already scholars have pointed to the emergence of nontraditional forms of large-scale ecological governance (for example, Wilson Rowe 2021), the distinctive (and frequently effective) climate diplomacy of small states (for example, Morgan, Carter and Manoa 2024), to forms of localised governance in response to climate effects (for example, McNaught 2024) and to Indigenous forms of ecosystem management embedded in relational approaches to the natural world that challenge the human-nature divide (for example, Reed et al 2021). Ongoing grassroots activists also consistently make a case for prioritizing attention to the most vulnerable and addressing the direct harms of climate change (see, for example, Garcia-Gibson 2023). Indeed, Aly Tkachenko (2024) makes the case that existing forms of grassroots direct action to protect local environments can be viewed as ecological security in practice.

This reminds us of two important points in considering the prospects for ecological security and the dynamic relationship between nature governance and global order. First, we cannot allow existing institutional arrangements and their capacity – or preferred policy instruments – to define the limits of our conception of politics, sites or means of response (on this point, see Lövbrand et al 2015). Second, addressing climate change necessitates action below and above those traditional security providers – the state and its defence establishment. In this context, recognizing the nature of the challenge that climate change poses means recognizing the potential – even the imperative – of new forms of governance.

Conclusion

Traditional approaches to security in IR are ill-equipped to come to terms with the nature of the Anthropocene or the ecological crises – the 'omni-crises' – we face. Nor do such approaches orient towards *directly* addressing the threats posed by ecological crises, most notably climate change. This necessitates an orientation towards ecological security, which recognizes direct and immediate harms experienced by the most vulnerable, and prioritizes responses to these harms. Traditional accounts, by contrast, focus on the preservation or protection of self-contained units, and their insulation

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from the effects of an ecological crisis like climate change. This might be consistent with the logics or pathologies of (dominant accounts of) security, but it sits uneasily with the nature of this crisis. As noted in the preceding analysis, such an approach fits poorly with recognition of the immediate and direct harms associated with climate change, with the range of means and array of actors necessary to address it and, perhaps most fundamentally, struggle to come to terms with the core question of who is rendered most fundamentally and immediately vulnerable to the effects of climate change and ecological crises more broadly.

As the preceding analysis has also made clear, the tendencies within traditional theoretical approaches to IR are replicated and reinforced by the practices of key institutions of global politics. Here, even when institutions like states and the UNSC engage directly with the security implications of climate change, there is an overwhelming commitment to make climate change fit within existing accounts of the remit and scope of the organization and the logics of security with which they are comfortable. The result is – at best – the pursuit of piecemeal, partial and predominantly adaptive responses oriented towards the survival of the institution itself, not the fundamental shift in orientation and practice required to address the climate crisis. As the onset of the climate crisis deepens and worsens, the failure of business-as-usual conceptions and practices of security in response to that crisis will increasingly raise questions about the legitimacy of these institutions and also potentially the contours of global order.

This *tendency* in orientation does not, as noted in the preceding pages, apply to all scholarship or all institutions. At the institutional level, aside from increasing engagement with the security implications of climate change in the UNSC (see Maertens 2021; McDonald 2023b) and among states (see Vogler 2023, 2024), we can also see at least some movements away from the model of the self-contained state insulating itself from the (indirect) effects of climate change among those institutions, as noted by policy practitioners. In IR scholarship generally and security scholarship specifically, we can also see increasing recognition of the climate crisis and the Anthropocene context, along with reflection on what this means for changing the way we think about security and how it might be advanced or realized. This is evident in the critical scholarship on security generally and scholars making the case for ecological security – or variants of it – more specifically. In this sense, recognition of the realities of dealing with the climate crisis might yet drive changes in modes of global governance and practice, along with the conceptual frameworks used to make sense of these dynamics and guide responses to them. And there is arguably a role for scholars and scholarship attempting to advance this case in public debate, in the corridors of power and in the academy, a project to which this book contributes.

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Notes

1 Although as Welsh (2022: 9) notes, it is more appropriate to position the referent object of security in this schema as the 'state' rather than the 'nation'.

2 Of course, for a critique of this 'logic' of anarchy – and the assumption that this compels self-help and mistrust – see Wendt (1992).

3 It should also be noted that the proposal to formally endorse the Anthropocene epoch, which dated from the 1950s, failed to receive a supermajority of votes from a panel of the International Commission on Stratigraphy in March 2024.

4 Indeed, Article 24 of the UN Charter notes that: 'In order to ensure prompt and effective action by the United Nations, its Members confer on the Security Council *primary responsibility* for the maintenance of international peace and security.'

5 The states of the Pacific Island Forum defined climate change primarily as a direct threat to human security in their 2018 Boe Declaration (PIF 2018), while policy statements of climate security in Sweden and Germany, for example, identify the immediate effects of climate change for vulnerable states in the developing world.

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